

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

May 2012

CL305 Environmental Engineering-II

Date: 04.05.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 (a) Describe environmental impact of waste generation. [05]
(b) Define the terms (any two) [02]
(1) sullage (2) refuse (3) sewerage (4) garbage
- Q-2 (a) Briefly explain various types of biological systems employed for treatment of domestic sewage. [07]
(b) Briefly describe the factors affecting dry weather flow of sewage. [04]
(c) Calculate population equivalent of a city given (i) the average sewage from city is 80×10^6 Lit/day, and (ii) the average 5-day BOD of sewage is 250 mg/L. Assume domestic sewage BOD load as 0.08 kg/capita/day. [03]

OR

- Q-2 (a) Define BOD. Derive expression for first stage BOD. [06]
(b) The 5-day BOD (20°C) of a sewage sample was found to be 100 mg/L. Calculate 2-day BOD at 30°C for the same sample. Assume $K_{20} = 0.1$ /day (base 10) [04]
(c) Design a manually cleaned bar screen for a peaked flow of 40 million liters per day. Clear spacing between bars may be taken as 25-50 mm and velocity of flow through the screen as 0.7-1 m/s. Assume suitable data and clearly mention. [04]
- Q-3 (a) Explain with a neat sketch biological process occurring in a trickling filter. [05]
(b) Explain self cleansing and non-scouring velocity in sewers. [05]
(c) Design a circular primary settling tank for a town having population of 50000 and sewage discharge rate of 144 lit/capita/day. [04]

OR

- Q-3 (a) Explain briefly various methods of aeration employed in activated sludge treatment. [05]
(b) Design a standard rate trickling filter for a sewage flow of 4.5 million liters per day [06]

and 5-day BOD of 160 mg/L. Assume organic loading as 160 gm/m³/day and surface loading of 2000 L/ m²/day. Also calculate efficiency of the filter using NRC equation.

- (c) Explain importance of recirculation in trickling filters. [03]

SECTION - II

- Q-4 (a) Describe various factors affecting quantity of municipal solid waste. [04]

- (b) Explain primary and secondary air pollutants giving few examples of each. [03]

- Q-5 (a) Describe with neat sketch the disposal of solid waste in a sanitary landfill. [07]

- (b) Describe briefly various effects of air pollution on vegetation. [04]

- (c) Explain the sources and characteristics of oxides of nitrogen. [03]

OR

- Q-5 (a) Describe with neat sketch working of a packed absorber tower. [04]

- (b) What is meant by smog? Discuss its causes and effects. [04]

- (c) With a neat sketch explain principle and working of an electrostatic precipitator. [06]

- Q-6 (a) Describe effect of sulfur dioxide and lead on human health [04]

- (b) Write a short note on Environmental Impact Assessment. [05]

- (c) Enlist various devices used for control of particulate air pollutants and describe briefly bag (or fabric) filter. [05]

OR

- Q-6 (a) Briefly describe composting used for municipal solid waste. [04]

- (b) What are the advantages and disadvantages of environmental auditing? [05]

- (c) Write a short note on cyclone used for particulate matter removal. [05]
